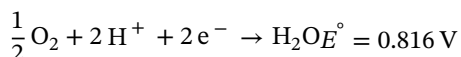


Catabolism is an oxidative process in which electrons are transferred from lipids, proteins, and carbohydrates to generate high-energy electron carriers from nicotinamide adenine dinucleotide (NADH/NAD⁺) and flavin adenine dinucleotide (FADH₂/FAD). Both are used in the electron transport chain (ETC) as electron providers in a series of reactions. The transfer of electrons provides energy that drives oxidative phosphorylation and the production of ATP. Reduction of NAD⁺ proceeds according to the half-reaction shown in Reaction 1.



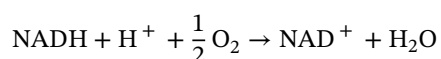
Reaction 1

Reactions in the ETC occur in four protein complexes (I–IV) located in the inner mitochondrial membrane. Here high-energy electrons are shuttled from complexes I, II, and III to complex IV, where O₂ serves as the final electron acceptor according to the half-reaction shown in Reaction 2.



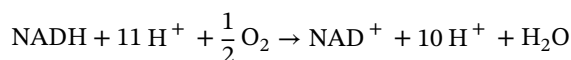
Reaction 2

The net result of NADH oxidation is described by Reaction 3.



Reaction 3

At complexes I, III, and IV, hydrogen ions move from the mitochondrial matrix to the intermembrane space and create the electrochemical gradient, or *proton motive force* (pmf), that drives ATP synthesis. Each NADH molecule drives the net pumping of 10 protons across the membrane, as per Reaction 4.



Reaction 4

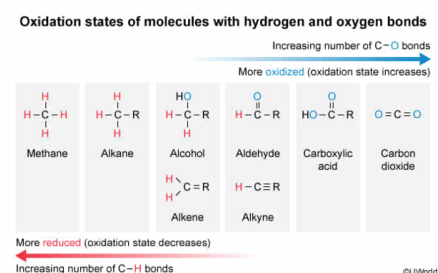
For biological processes, which of the following is true of oxidation-reduction reactions?

- I. Reduction often involves the formation of bonds to hydrogen.
 - II. Oxidation often involves the formation of bonds to oxygen.
 - III. Reducing agents undergo a change to a lower oxidation state.
- ☐ A. I only [11%]
☒ B. I and II only [62%]
☐ C. II and III only [5%]
☐ D. I, II, and III [20%]

Correct

62% answered correctly

Explanation:



Oxidation-reduction (redox) reactions are reactions in which electrons are transferred from one molecule to another. One way to remember the flow of electrons in a redox reaction is with the acronym "OIL RIG": **oxidation** is **loss** of electrons and **reduction** is **gain** of electrons. However, different compounds transfer electrons in different ways. Unlike

metals, which transfer electrons directly and have an oxidation state equal to their net charge, organic molecules generally transfer electrons via hydrogen or oxygen.

When **hydrogen forms a bond** to carbon, the more **electronegative** carbon atom takes the electron from hydrogen and is **reduced (Number I)**. The loss of a bond to hydrogen or formation of a **bond to electronegative oxygen** results in carbon losing electrons, which is **oxidation (Number II)**.

A simple way to determine the oxidation state of an organic molecule is to count the number of C–O and C–H bonds. For example, an **aldehyde** is in a higher oxidation state than an **alcohol** because it has two C–O bonds in the form of a double bond compared to the single C–O bond in alcohols. Alkanes are more reduced than alkenes because they have more bonds to hydrogen.

(Number III) **Reducing agents** cause the reduction (lowers the oxidation state) of other molecules, but a reducing agent itself gets oxidized in the process, which gives it a *higher* oxidation state.

Educational objective:

For organic molecules, oxidation involves the loss of electrons by forming bonds to oxygen or losing bonds to hydrogen; reduction is the opposite process. The compound that becomes oxidized is the reducing agent, and the compound that becomes reduced is the oxidizing agent.

Time spent:0 sec
Subject:
General Chemistry

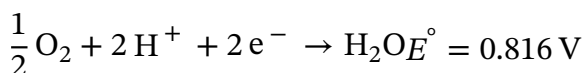
QID:400335
System:
Solutions and Electrochemistry

Catabolism is an oxidative process in which electrons are transferred from lipids, proteins, and carbohydrates to generate high-energy electron carriers from nicotinamide adenine dinucleotide (NADH/NAD⁺) and flavin adenine dinucleotide (FADH₂/FAD). Both are used in the electron transport chain (ETC) as electron providers in a series of reactions. The transfer of electrons provides energy that drives oxidative phosphorylation and the production of ATP. Reduction of NAD⁺ proceeds according to the half-reaction shown in Reaction 1.



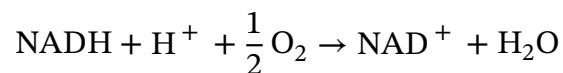
Reaction 1

Reactions in the ETC occur in four protein complexes (I–IV) located in the inner mitochondrial membrane. Here high-energy electrons are shuttled from complexes I, II, and III to complex IV, where O₂ serves as the final electron acceptor according to the half-reaction shown in Reaction 2.



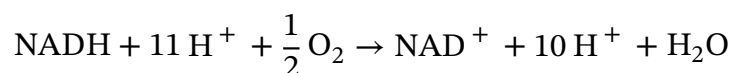
Reaction 2

The net result of NADH oxidation is described by Reaction 3.



Reaction 3

At complexes I, III, and IV, hydrogen ions move from the mitochondrial matrix to the intermembrane space and create the electrochemical gradient, or *proton motive force* (pmf), that drives ATP synthesis. Each NADH molecule drives the net pumping of 10 protons across the membrane, as per Reaction 4.



Reaction 4

What is the net reaction potential under standard conditions for the overall reaction that occurs between NADH and O₂ within the ETC?

- ☐ A. -1.136 V [9%]
- ☐ B. -0.496 V [4%]
- ☐ C. 0.496 V [24%]
- ☒ D. 1.136 V [62%]

Correct

61% answered correctly

Explanation:

Determining the net standard reduction potential (E°) of a redox reaction		
Half-reaction 1 (Reduction)	$\text{NAD}^+ + 2\text{H}^+ + 2\text{e}^- \rightarrow \text{NADH} + \text{H}^+$	$E^\circ_{\text{Reduction}} = -0.320\text{ V}$
↓		Flipped sign ↓
Reversed half-reaction 1 (Oxidation)	$\text{NADH} + \text{H}^+ \rightarrow \text{NAD}^+ + 2\text{H}^+ + 2\text{e}^-$	$E^\circ_{\text{Oxidation}} = +0.320\text{ V}$
Half-reaction 2 (Reduction)	$+\frac{1}{2}\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2\text{O}$	$E^\circ_{\text{Reduction}} = +0.816\text{ V}$
Net reaction	$\text{NADH} + \text{H}^+ + \frac{1}{2}\text{O}_2 \rightarrow \text{NAD}^+ + \text{H}_2\text{O}$	$E^\circ_{\text{Net reaction}} = +1.136\text{ V}$

Redox reactions occur when one or more electrons are transferred from one atom to another. The atom that **loses** electrons is **oxidized** (ie, electron donor) and the atom that **gains** electrons is **reduced** (ie, electron acceptor). Redox reactions are composed of a reduction half-reaction and an oxidation half-reaction. The net **standard potential** for a redox reaction ($E^\circ_{\text{net reaction}}$) is equal to the sum of E° for each half-reaction:

$$E^\circ_{\text{net reaction}} = E^\circ_{\text{reduction}} + E^\circ_{\text{oxidation}}$$

Because half-reactions are typically written as reductions, one of the half-reactions must be reversed to make it oxidation, and thus the **sign of its E° value is flipped**.

The passage states that O_2 is the final electron acceptor (ie, O_2 is reduced). Therefore, the net reaction potential is calculated as the sum of the standard reduction potential of O_2 (+0.816 V) and the standard oxidation potential of NADH (+0.320 V):

$$E^\circ_{\text{net reaction}} = 0.816\text{ V} + 0.320\text{ V} = 1.136\text{ V}$$

(Choice A) This value, which is the correct magnitude but the wrong sign, is the potential for the reverse reaction in which electrons are transferred from H_2O to NAD^+ .

(Choices B) Erroneously flipping the equations and corresponding signs of E° for both half-reactions results in an answer of -0.496 V .

(Choice C) Adding the given reduction potentials for Reactions 1 and 2 without flipping the sign of E° for Reaction 1 results in an answer of 0.496 V .

Educational objective:

The standard electric potential of a redox reaction is the sum of the standard potentials of the oxidation and reduction half-reactions.

Time spent: 0 sec

Subject:
General Chemistry

QID: 400336

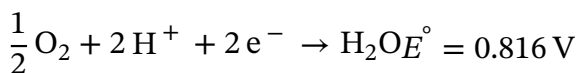
System:
Solutions and Electrochemistry

Catabolism is an oxidative process in which electrons are transferred from lipids, proteins, and carbohydrates to generate high-energy electron carriers from nicotinamide adenine dinucleotide (NADH/NAD⁺) and flavin adenine dinucleotide (FADH₂/FAD). Both are used in the electron transport chain (ETC) as electron providers in a series of reactions. The transfer of electrons provides energy that drives oxidative phosphorylation and the production of ATP. Reduction of NAD⁺ proceeds according to the half-reaction shown in Reaction 1.



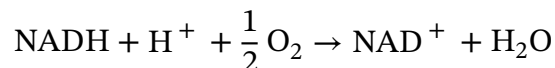
Reaction 1

Reactions in the ETC occur in four protein complexes (I–IV) located in the inner mitochondrial membrane. Here high-energy electrons are shuttled from complexes I, II, and III to complex IV, where O₂ serves as the final electron acceptor according to the half-reaction shown in Reaction 2.



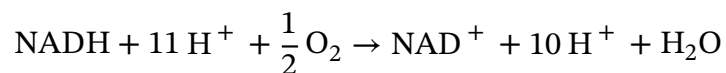
Reaction 2

The net result of NADH oxidation is described by Reaction 3.



Reaction 3

At complexes I, III, and IV, hydrogen ions move from the mitochondrial matrix to the intermembrane space and create the electrochemical gradient, or *proton motive force* (pmf), that drives ATP synthesis. Each NADH molecule drives the net pumping of 10 protons across the membrane, as per Reaction 4.



Reaction 4

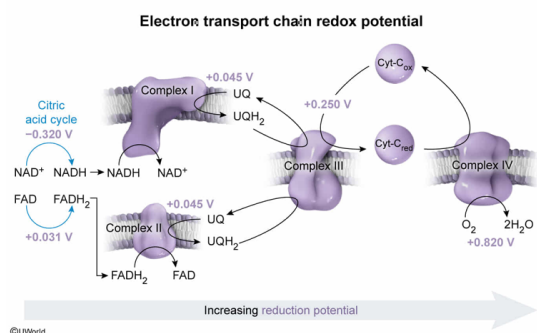
If the ETC is a spontaneous process, which of the following **MUST** be true about the redox reactions within each of the four complexes?

- ☐ A. The electron-carrying molecules in the ETC have E° values greater than that of oxygen. [16%]
- ☐ B. The overall free energy ΔG° of the ETC is positive. [3%]
- ☒ C. The E° values increase as electrons are passed from one carrier to the next. [75%]
- ☐ D. The redox reactions in complex IV are faster than the reactions in complex I. [4%]

Correct

76% answered correctly

Explanation:



Standard reduction potentials (E°) are indicative of whether a molecule will **spontaneously gain or lose electrons**. In a spontaneous process, E° is positive. The ETC is a series of spontaneous processes, so all the standard reduction potentials must fall within the range of NAD^+ ($E^\circ = -0.320 \text{ V}$) and O_2 ($E^\circ = 0.816 \text{ V}$), and each step in the chain must have a larger E° than the previous step.

(Choice A) The reduction potentials for each electron carrier must be less than that of oxygen for the process to be spontaneous.

(Choice B) For a process to be spontaneous, the Gibbs free energy ΔG° must be negative. Gibbs free energy is inversely proportional to standard reduction potential, as described by the equation, $\Delta G^\circ = -nFE^\circ$.

(Choice D) The reactions in the ETC move spontaneously from one complex to the next. However, the spontaneity of a reaction does not provide a direct correlation to the rate of a reaction.

Educational objective:

Standard reduction potentials (E°) indicate whether a molecule spontaneously gains or loses electrons and are related to the Gibbs free energy (ΔG°). In a spontaneous process, E° is positive, ΔG° is negative, and electrons flow from molecules with low reduction potential to molecules with high reduction potential.

Time spent: 0 sec

Subject:
General Chemistry

QID: 400337

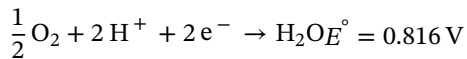
System:
Solutions and Electrochemistry

Catabolism is an oxidative process in which electrons are transferred from lipids, proteins, and carbohydrates to generate high-energy electron carriers from nicotinamide adenine dinucleotide (NADH/NAD⁺) and flavin adenine dinucleotide (FADH₂/FAD). Both are used in the electron transport chain (ETC) as electron providers in a series of reactions. The transfer of electrons provides energy that drives oxidative phosphorylation and the production of ATP. Reduction of NAD⁺ proceeds according to the half-reaction shown in Reaction 1.



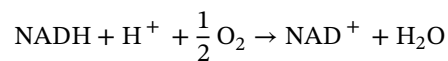
Reaction 1

Reactions in the ETC occur in four protein complexes (I–IV) located in the inner mitochondrial membrane. Here high-energy electrons are shuttled from complexes I, II, and III to complex IV, where O₂ serves as the final electron acceptor according to the half-reaction shown in Reaction 2.



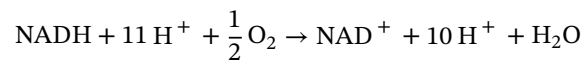
Reaction 2

The net result of NADH oxidation is described by Reaction 3.



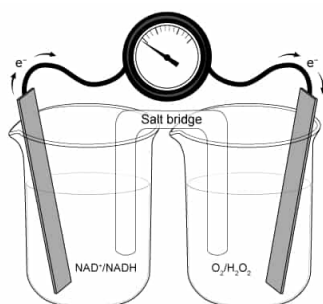
Reaction 3

At complexes I, III, and IV, hydrogen ions move from the mitochondrial matrix to the intermembrane space and create the electrochemical gradient, or *proton motive force* (pmf), that drives ATP synthesis. Each NADH molecule drives the net pumping of 10 protons across the membrane, as per Reaction 4.



Reaction 4

Researchers want to measure the reduction potential when NADH is oxidized by oxygen to compare the experimental reduction potential with the calculated reduction potential in the mitochondria. A galvanic cell was constructed as shown in the following diagram.



Which of the following statements is true about NAD⁺/NADH in this experiment?

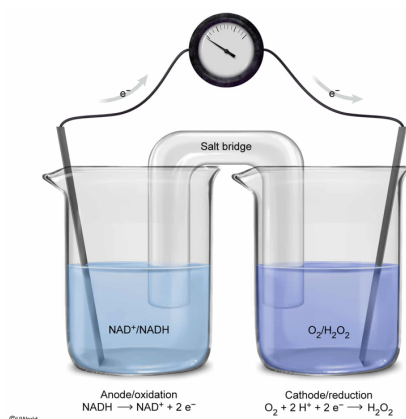
- ☐ A. NAD⁺ becomes oxidized at the anode. [7%]
- ☐ B. An applied potential causes NADH to lose electrons at the cathode. [5%]

- ☐ C. NADH is oxidized at the anode, and NAD^+ is reduced at the cathode. [36%]
- ✔ ☒ D. An electric potential exists because NADH loses electrons at the anode. [51%]

Correct

51% answered correctly

Explanation:



In a **galvanic cell**, a **spontaneous** ($\Delta G^\circ < 0$) **reduction-oxidation** (redox) reaction occurs, producing a flow of electrons between two electrodes connected by a wire. Electrons are generated at the electrode called the **anode** (ie, where **oxidation** occurs) and flow through the wire toward the electrode called the **cathode** (ie, where **reduction** occurs).

According to the question, NADH is oxidized by O_2 ; therefore, NADH is oxidized to NAD^+ at the anode. This process produces an electric potential as NADH spontaneously generates electrons that flow from the anode to the cathode.

(Choice A) Although oxidation occurs at the anode, NAD^+ is the product of NADH oxidation and does not itself become oxidized.

(Choice B) An electric potential already exists because NADH spontaneously loses two electrons to form NAD^+ ; no potential needs to be applied. Furthermore, the loss of electrons occurs at the anode rather than the cathode.

(Choice C) Galvanic cells are composed of two half-cells, with one half containing the molecule that becomes oxidized at the anode (in this case, NADH) and the other half containing the molecule that becomes reduced at the cathode (in this case, O_2). NADH and NAD^+ must be in the same half-cell because one is made from the other. Similarly, O_2 and H_2O_2 must be together in the other half-cell.

Educational objective:

Galvanic cells are electrochemical cells that produce an electric potential as a result of the spontaneous transfer of electrons from the anode to the cathode. In any electrochemical cell, oxidation occurs at the anode and reduction occurs at the cathode.

Time spent: 0 sec
Subject:
General Chemistry

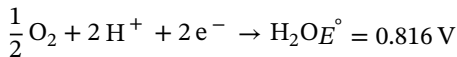
QID: 400338
System:
Solutions and Electrochemistry

Catabolism is an oxidative process in which electrons are transferred from lipids, proteins, and carbohydrates to generate high-energy electron carriers from nicotinamide adenine dinucleotide (NADH/NAD⁺) and flavin adenine dinucleotide (FADH₂/FAD). Both are used in the electron transport chain (ETC) as electron providers in a series of reactions. The transfer of electrons provides energy that drives oxidative phosphorylation and the production of ATP. Reduction of NAD⁺ proceeds according to the half-reaction shown in Reaction 1.



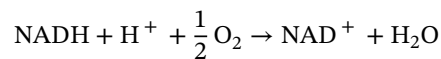
Reaction 1

Reactions in the ETC occur in four protein complexes (I–IV) located in the inner mitochondrial membrane. Here high-energy electrons are shuttled from complexes I, II, and III to complex IV, where O₂ serves as the final electron acceptor according to the half-reaction shown in Reaction 2.



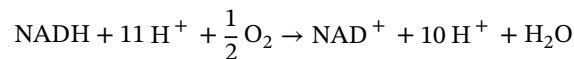
Reaction 2

The net result of NADH oxidation is described by Reaction 3.



Reaction 3

At complexes I, III, and IV, hydrogen ions move from the mitochondrial matrix to the intermembrane space and create the electrochemical gradient, or *proton motive force* (pmf), that drives ATP synthesis. Each NADH molecule drives the net pumping of 10 protons across the membrane, as per Reaction 4.



Reaction 4

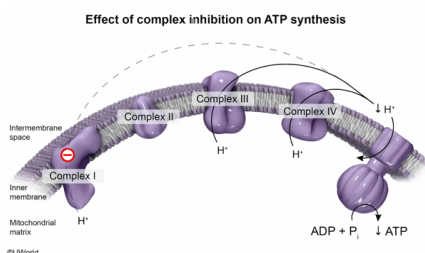
Barbiturates are known to inhibit complex I but do not affect complexes II–IV. Based on this fact, how will barbiturates affect the proton motive force?

- ☐ A. The proton motive force increases. [2%]
- ☒ B. The proton motive force decreases. [92%]
- ☐ C. The proton motive force is unchanged. [4%]
- ☐ D. The effect cannot be determined without more information. [1%]

Correct

91% answered correctly

Explanation:



As stated in the passage, the proton motive force (pmf) arises as hydrogen ions (protons) are pumped from the mitochondrial matrix into the intermembrane space by complexes I, III, and IV. It is the driving force behind ATP synthesis in the mitochondria. When complex I is **inhibited**, the ETC can still operate from complex II onward as long as FADH₂ can continue to provide complex II with electrons.

Because complex I normally participates in proton pumping, its inhibition will result in fewer protons being pumped across the mitochondrial membrane. Therefore, the **H⁺ concentration gradient is reduced** and the **pmf decreases**,

resulting in decreased ATP synthesis.

(Choice A) An increase in the pmf would require increased proton pumping. Shutting down complex I reduces proton pumping.

(Choices C and D) Because complex I is involved in proton pumping, the pmf must change when the complex is blocked. Because fewer protons are pumped when complex I is blocked, the pmf must decrease. Therefore, the question and passage give sufficient information to determine the effect of barbiturates on the pmf.

Educational objective:

Inhibition of any complex in the ETC results in fewer protons being pumped into the intermembrane space of the mitochondria, resulting in a decreased proton motive force and decreased ATP synthesis.

Time spent:0 sec

Subject:
General Chemistry

QID:400339

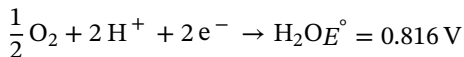
System:
Solutions and Electrochemistry

Catabolism is an oxidative process in which electrons are transferred from lipids, proteins, and carbohydrates to generate high-energy electron carriers from nicotinamide adenine dinucleotide (NADH/NAD⁺) and flavin adenine dinucleotide (FADH₂/FAD). Both are used in the electron transport chain (ETC) as electron providers in a series of reactions. The transfer of electrons provides energy that drives oxidative phosphorylation and the production of ATP. Reduction of NAD⁺ proceeds according to the half-reaction shown in Reaction 1.



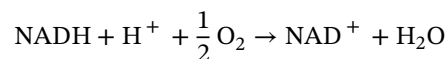
Reaction 1

Reactions in the ETC occur in four protein complexes (I–IV) located in the inner mitochondrial membrane. Here high-energy electrons are shuttled from complexes I, II, and III to complex IV, where O₂ serves as the final electron acceptor according to the half-reaction shown in Reaction 2.



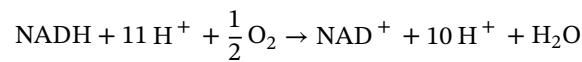
Reaction 2

The net result of NADH oxidation is described by Reaction 3.



Reaction 3

At complexes I, III, and IV, hydrogen ions move from the mitochondrial matrix to the intermembrane space and create the electrochemical gradient, or *proton motive force* (pmf), that drives ATP synthesis. Each NADH molecule drives the net pumping of 10 protons across the membrane, as per Reaction 4.



Reaction 4

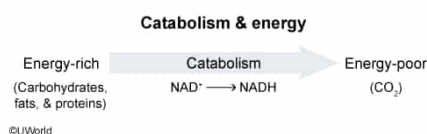
Which of the following is NOT true concerning catabolism and the function of the ETC?

- ☐ A. The number of ATP molecules produced is dependent on the ΔG° of the ETC. [17%]
- ☐ B. ΔG° of the ETC depends on the overall E° of the ETC. [8%]
- ☒ C. Catabolism requires NADH to serve as an oxidizing agent. [64%]
- ☐ D. Catabolism produces energy in the form of electron transduction. [9%]

Correct

65% answered correctly

Explanation:



Catabolism breaks down biological fuel molecules such as carbohydrates, lipids, and proteins, releasing energy as they are **oxidized**. Oxidation cannot occur without reduction; consequently, a different molecule must be reduced in the process.

NADH can be oxidized but not reduced, and therefore it serves as a **reducing agent**, not an oxidizing agent. Instead, NAD⁺ serves as an oxidizing agent for catabolic reactions and is reduced to NADH in the process. NADH then enters the ETC as a reducing agent *after* oxidation of fuel molecules, and thus provides the energy necessary for ATP production.

The question asks which of the given statements is *not* true concerning catabolism and the function of the ETC.

Choice C provides the correct answer because NADH does *not* serve as an oxidizing agent in catabolism.

(Choice A) ATP production requires energy input, which is measured by ΔG° . The number of ATP molecules that can be produced depends on the energy available.

(Choice B) ΔG° (Gibbs free energy) depends on E° (reduction potential) and is related by the relationship: $\Delta G^\circ = -nFE^\circ$.

(Choice D) NADH and FADH_2 , which are produced by catabolic oxidation of lipids, proteins, and carbohydrates, provide energy for ATP production by passing high-energy electrons through the ETC. This process is also known as electron transduction.

Educational objective:

Catabolism is an oxidative process that releases energy. During catabolic oxidation of fuel molecules, oxidizing agents such as NAD^+ are reduced. Electron transduction from these reduced cofactors provides the energy for ATP synthesis.

Time spent: 0 sec

Subject:
General Chemistry

QID: 400340

System:
Solutions and Electrochemistry